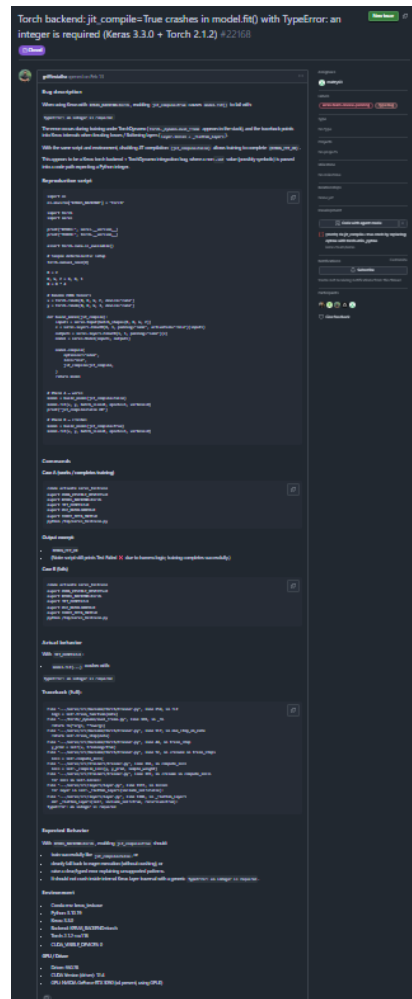


Keras Reported Bugs:

Repo Link:

[Torch backend: jit_compile=True crashes in model.fit\(\) with TypeError: an integer is required \(Keras 3.3.0 + Torch 2.1.2\) · Issue #22168 · keras-team/keras](https://github.com/keras-team/keras/issues/22168)



Repo Link:

[Passing a plain callable to add_weight\(constraint=...\) raises ValueError in Keras 3 \(torch backend\) · Issue #22221 · keras-team/keras](#)

[illegible]

Repo Link:

[Layer.add_weight\(\) misinterprets positional first argument as shape, raising "got multiple values for argument shape". · Issue #22253 · keras-team/keras](#)

[illegible]

Repo Link:

[keras_cv.TextEncoder: predict on batch output differs from direct call \(TensorFlow backend, GPU\) · Issue #2620 · keras-team/keras-hub](#)

keras_cv.TextEncoder.predict_on_batch output differs from direct call (TensorFlow backend, GPU) #2620

Repo Link:

[CLIPBackbone text_encoder: predict_on_batch\(\) output differs from tf.function on GPU \(TF backend\) · Issue #22380 · keras-team/keras](#)

CLIPBackbone text_encoder: predict_on_batch() output differs from tf.function on GPU (TF backend) #22380

Open

gR0nd4ke opened 1 month ago

Describe the bug

Using tf.nn.softmax() on GPU (tf.function) the text encoder produces different outputs between:

- `model.predict_on_batch(x_test_data)`
- `model.predict(x_test_data)`

for the same model weights and same input data. This is not a bug in the model, but in the `tf.nn.softmax()` function.

Expected behavior

The outputs of `model.predict_on_batch()` and `model.predict(x_test_data)` should be identical on GPU for the same model weights and input data.

To Reproduce

```
import numpy as np
import tensorflow as tf

# Create a simple model
model = tf.keras.Sequential([
    tf.keras.layers.Dense(10, activation='softmax')
])

# Create some test data
x_test_data = np.random.randn(100, 10)

# Predict using model.predict_on_batch()
y_pred_on_batch = model.predict_on_batch(x_test_data)

# Predict using model.predict()
y_pred_predict = model.predict(x_test_data)

# Compare the outputs
print(np.allclose(y_pred_on_batch, y_pred_predict))
```

Expected output

The output of `np.allclose(y_pred_on_batch, y_pred_predict)` should be `True`.

Actual output

The output of `np.allclose(y_pred_on_batch, y_pred_predict)` is `False`.

Environment

OS: Ubuntu 20.04.2 LTS
Python: 3.8.10
TensorFlow: 2.4.0
Keras: 2.4.3

Deepspeed Reported Bugs:

Repo Link:

[\[BUG\] DeepSpeedEngine del /destroy throws AttributeError after initialize\(\) failure \(missing deepcompile active\) + BrokenPipeError noise · Issue #7812 · deepspeedai/DeepSpeed](#)

[illegible]

Repo Link:

[\[BUG\] ZeRO-3: zero.GatheredParameters\(\[multiple params\], modifier rank=None\) + in-place slice touch triggers assert not param.ds_active_sub_modules in free_param\(\) · Issue #7811 · deepspeedai/DeepSpeed](#)

[illegible]

Repo Link:

[\[BUG\] DistributedAttention produces numerically incorrect results vs its own SeqAllToAll-equivalent baseline when sequence length is not divisible by world size · Issue #7842 · deepspeedai/DeepSpeed](#)

[illegible]

Repo Link:

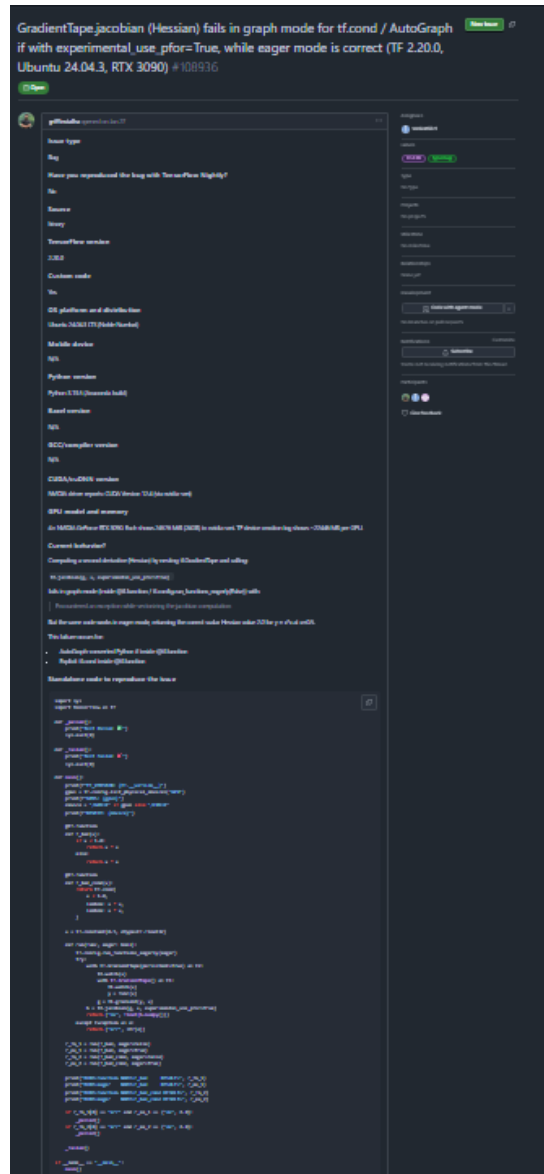
[\[BUG\] DeepSpeed ZeRO-3 deadlock in engine.step\(\) at step 0 under 2-GPU execution \(RTX 3090, torch 2.2.1, DS 0.14.2\) · Issue #7844 · deepspeedai/DeepSpeed](#)

[illegible]

Tensorflow Reported Bugs:

Repo Link:

[GradientTape.jacobian \(Hessian\) fails in graph mode for tf.cond / AutoGraph if with experimental_use_pfor=True, while eager mode is correct \(TF 2.20.0, Ubuntu 24.04.3, RTX 3090\) · Issue #108936 · tensorflow/tensorflow](#)



Repo Link:

[Keras model.fit\(\) crashes with ValueError: as_list\(\) is not defined on an unknown TensorShape when dataset elements come from tf.numpy_function without shapes · Issue #109333 · tensorflow/tensorflow](#)

[illegible]

Repo Link:

[Documentation mismatch: Layer/Model.add_metric\(\) shown usable in call\(\), but Keras 3 disables / breaks it \(TF 2.17 + Keras 3.4\) · Issue #110884 · tensorflow/tensorflow](#)

Documentation mismatch: Layer/Model.add_metric() shown usable in call(), but Keras 3 disables / breaks it (TF 2.17 + Keras 3.4) #110864

Repo Link:

[tf.math.reduce * accepts non-bool integers for keepdims, while comparable bool parameters in other tf.math APIs raise TypeError · Issue #112808 · tensorflow/tensorflow](#)

[illegible]

Repo Link:

[quantize_model\(\) rejects a valid standalone Keras 3 Sequential model · Issue #1270 · tensorflow/tensorflow](https://github.com/tensorflow/tensorflow/issues/1270)

quantize_model() rejects a valid standalone Keras 3 Sequential model #1270

Open

giffordalme

opened this issue on Jan 10, 2020

Describe the bug

`tf.keras.quantization.quantize_model()` rejects a valid standalone `tf.keras.Sequential` model with `tf.keras.layers.Dense` layers, but only when the model is not a `tf.keras.Sequential` model.

This appears to be a compatibility check between `tf.keras` and `tf.keras.Model` (subclass of `tf.keras.Sequential`). The model is a valid `tf.keras.Sequential` model, but `tf.keras` does not recognize it as a compatible `tf.keras.Sequential` model.

TensorFlow version: 2.2.0 (installed from binary / pip wheel)

Python version: 3.7.4

OS: Linux

Describe the expected behavior

A valid standalone `tf.keras.Sequential` model should be accepted by `tf.keras.quantization.quantize_model()` on the CPU/GPU. The model is a valid `tf.keras.Sequential` model, but `tf.keras` does not recognize it as a compatible `tf.keras.Sequential` model.

Describe the actual behavior

A standalone `tf.keras.Sequential` model is created successfully, but calling `tf.keras.quantization.quantize_model()` results in an error: `ValueError: Invalid argument: not a valid tf.keras.Sequential model.`

Code to reproduce the issue

```
import tensorflow as tf
import numpy as np
import tensorflow.keras as keras

model = tf.keras.Sequential()
model.add(keras.layers.Dense(10, activation='tanh'))
model.add(keras.layers.Dense(10, activation='tanh'))

x = np.random.randn(1, 10).astype(np.float32)
y = np.random.randn(1, 10).astype(np.float32)

model.compile(optimizer='adam', loss='mse')
model.fit(x, y, epochs=10)

quantized_model = tf.keras.quantization.quantize_model(model)
print(quantized_model)
```

Expected Output

```
tf.keras.Sequential(
  (Dense_1) Dense(10, activation=tanh)
  (Dense_2) Dense(10, activation=tanh)
)
Model: "tf.keras.Sequential"
Layer (type) Output Shape: (1, 10)
Dense_1 (Dense) (1, 10)
Dense_2 (Dense) (1, 10)
Total params: 20
Trainable params: 20
Non-trainable params: 0
Total size: 160.0B
Trainable size: 160.0B
```

Reproduction

Steps to reproduce the issue:

1. Create a standalone `tf.keras.Sequential` model.
2. Call `tf.keras.quantization.quantize_model()` on the model.
3. An error is raised: `ValueError: Invalid argument: not a valid tf.keras.Sequential model.`

Additional context

This does not appear to be a CPU/GPU issue. The same behavior occurs on both CPU and GPU.

My environment details

```
tf.__version__ = 2.2.0
tf.keras.__version__ = 2.2.0
tf.keras.backend.__version__ = 2.2.0
tf.keras.layers.__version__ = 2.2.0
tf.keras.models.__version__ = 2.2.0
tf.keras.optimizers.__version__ = 2.2.0
tf.keras.utils.__version__ = 2.2.0
```

Other observed that the model is recognized as a standalone `tf.keras.Sequential` model, but not as a `tf.keras` model. This suggests a compatibility check between `tf.keras` and `tf.keras.Model` (subclass of `tf.keras.Sequential`) is failing.

Notes for the issue

This looks like a bug in `tf.keras` type checking rather than a model construction bug in `tf.keras`.

More details about my environment

```
tf.__version__ = 2.2.0
tf.keras.__version__ = 2.2.0
tf.keras.backend.__version__ = 2.2.0
tf.keras.layers.__version__ = 2.2.0
tf.keras.models.__version__ = 2.2.0
tf.keras.optimizers.__version__ = 2.2.0
tf.keras.utils.__version__ = 2.2.0
```

giffordalme

added 1 label

Add a comment

Pytorch Reported Bugs:

Repo Link:

[torch.compile / Inductor generates invalid C++ \(undeclared zuf0/zuf1\) during compile \(CppCompileError\) on RTX 3090 · Issue #175057 · pytorch/pytorch](#)

torch.compile / Inductor generates invalid C++ (undeclared zuf0/zuf1) during compile (CxxCompileError) on RTX 3090 #175057